Topic C1: Particles

C1.1 The particle model

Summary

This short section introduces the particle model and its explanation of different states of matter. A simple particle model can be used to represent the arrangement of particles in the different states of matter and to explain observations during changes in state. It does not, however, explain why different materials have different properties. This explanation is that the particles themselves and how they are held together must be different in some way. Elements are substances that are made up of only one type of atom and atoms of different elements can combine to make compounds.

Underlying knowledge and understanding

Learners should be familiar with the different states of matter and their properties. Learners should be aware of the energy changes when a change of state occurs. They should also be familiar with changes of state in terms of the particle model. Learners should have sufficient grounding in the particle model to be able to apply it to unfamiliar materials and contexts.

Common misconceptions

Learners commonly intuitively adhere to the idea that matter is continuous. For example, they believe that the space between gas particles is filled or non-existent, or that particles expand when they are heated. The notion that empty space exists between particles is problematic because this lacks supporting sensory evidence. They also show difficulty understanding the concept of changes in state being reversible; this should be addressed during the teaching of this topic.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM1.1i	represent three-dimensional shapes in two dimensions and vice versa when looking at chemical structures, e.g. allotropes of carbon	M5b

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	**To include**	**Maths**	**Working scientifically**	
C1.1a describe the main features of the particle model in terms of states of matter and change of state		M5b	WS1.1a, WS1.1b	
C1.1b explain in terms of the particle model the distinction between physical changes and chemical changes				
C1.1c explain the limitations of the particle model in relation to changes of state when particles are represented by inelastic spheres (e.g. like bowling balls)	**that it does not take into account the forces of attraction between particles, the size of particles and the space between them**	M5b	WS1.1c	Observations of change of state with comparison to chemical changes.